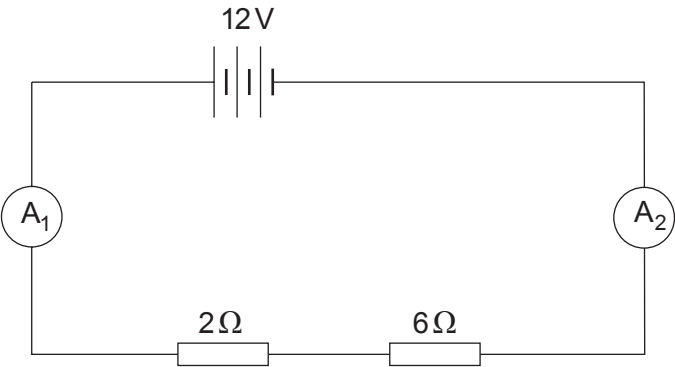


2. In class, a group of students set up a simple series circuit as shown in the diagram below.



(a) (i) Use the equation:

$$R = R_1 + R_2$$

to calculate the total resistance in the circuit. [1]

Total resistance = Ω

(ii) Use an equation from page 2 to calculate the current reading on ammeter A_1 . [2]

Ammeter A_1 reading = A

(iii) State the current reading on ammeter A_2 . [1]

Ammeter A_2 reading = A

(b) The 6Ω resistor is now removed from the circuit. It is added **in parallel** with the 2Ω resistor. The circuit is complete.

Complete the following sentences by underlining the correct phrase or word. [2]

- (i) The total resistance of the circuit (**increases** / **stays the same** / **decreases**).
- (ii) The current reading on the ammeter A_1 (**increases** / **stays the same** / **decreases**).

